

Lecture - 21

$Ax \rightarrow$ operation on vector x

$\Rightarrow$ Eigenvectors are vectors where Ax comes out as parallel to x .

$$Ax = \underset{\substack{\downarrow \\ \text{eigen value}}}{\lambda} x$$

If A is singular, $\lambda = 0$ is an eigenvalue.

What are the x 's & λ 's for the projection matrix.

→ Any vector x in the projection space will have Px parallel to itself and will be an eigenvector.

$$Px = x$$

$$\underline{\underline{\lambda = 1}}$$

→ Also any vector $x \perp$ to the projection space is also an eigenvector

$$\underline{\underline{Px = 0}}$$

$$\underline{\underline{\lambda = 0}}$$

$\lambda = 0, 1$ for projection matrices!

What about permutation matrices $\rightarrow$

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\# \quad x = c \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \boxed{\lambda = 1}$$

$$\# \quad x = c \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \boxed{\lambda = -1}$$

Fact

$$\boxed{\sum_{i=1}^n \lambda_i = \text{trace}(A)}$$

For $n \times n$ matrix, there are n eigenvalues.

⇒ Now to solve $Ax = \lambda x$

$$\rightarrow (A - \lambda I)x = 0$$

Now we can see that for eigen vectors to exist, the above matrix $A - \lambda I$ must be singular otherwise $x = 0$ would be the only solution.

$$\rightarrow \det(A - \lambda I) = 0$$

↳ After we have found λ , then we can just solve for nullspace and find the solutions.

eg $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{bmatrix}$$

$$(3-\lambda)^2 - 1^2 = 0$$

$$(3-\lambda+1)(3-\lambda-1) = 0$$

$$(4-\lambda)(2-\lambda) = 0$$

$$\boxed{\lambda = 2, 4}$$

$$A - \lambda I \rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\boxed{x \rightarrow \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

Same eigenvectors as $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^{(A_0)}$ $A = A_0 + 3I$

Neat fact

⇒ If we add cI to a matrix,
it's eigenvectors stay the same &
it's eigenvalues increase by c .

⇒ Suppose we have a matrix
if $Ax = \lambda x$ $Bx = \alpha x$

what about $(A+B)x = (\lambda + \alpha)x$

→ This is false as x might be
 y , they might have different
eigenvectors.

Q

$$Q = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\text{trace} = \lambda_1 + \lambda_2 = 0$$

$$\det Q = \lambda_1 \lambda_2 = 1$$

$$\det(Q - \lambda I) \Rightarrow \lambda^2 + 1 = 0$$

$$\lambda = \pm i \quad \pm \sqrt{-1}$$

→ Since this is a rotation matrix of 90° , there is no way a vector can come out parallel to itself.

- This is seen from the fact that its eigen values are imaginary numbers or complex terms.
- They come in complex conjugate pairs.

⇒ There is some relation between symmetry of matrices & its eigenvalues being real.
(Is there a tipping point?!))

eg $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$

$$\lambda_1 + \lambda_2 = 6$$

$$\lambda_1 \cdot \lambda_2 = 9$$

$$\boxed{\lambda_1 = \lambda_2 = 3}$$

For triangular matrices eigenvalues are on the diagonal!

⇒ They have linearly dependent eigenvectors.

